PROJECT DOCUMENTATION

STREAMING TICKET ASSIGNMENT FOR EFFICIENT SUPPORT OPERATIONS

Submitted By

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1.ABSTRACT

This project focuses on improving the efficiency of support operations by automating the assignment of support tickets using the ServiceNow platform. In large organizations, support teams often handle a high volume of tickets related to different services such as platform access issues, login problems, certificate requests, system errors, and other operational concerns. When these tickets are assigned manually, it can lead to delays in issue resolution, incorrect routing of tickets, increased workload for support staff, and reduced customer satisfaction.

To address these challenges, the project proposes an automated ticket routing system that intelligently assigns support tickets to the appropriate support groups based on the type of issue reported. The system is implemented using various features of the ServiceNow platform, including Users, Groups, Roles, Custom Tables, Access Control Lists (ACL), and Flow Designer. These components work together to create a structured environment where tickets are automatically analyzed and routed to the correct support team without manual intervention.

The project involves creating dedicated support groups for handling different categories of issues, such as platform-related problems and certificate-related requests. Automated workflows are then designed using Flow Designer to monitor newly created or updated records in the system. Based on predefined conditions, the workflows automatically assign tickets to the relevant support groups, ensuring faster and more accurate ticket distribution.

By implementing this automated approach, the organization can significantly reduce manual effort, improve response time, and ensure that issues are handled by the appropriate teams. The solution enhances operational efficiency, optimizes resource utilization, and improves the overall customer support experience by ensuring timely and effective resolution of support requests.

2.INTRODUCTION

2.1 Overview of ServiceNow

ServiceNow is a cloud-based platform that helps organizations automate business processes and manage IT services efficiently. It provides a centralized system for handling incidents, service requests, problems, and other support operations. ServiceNow is widely used in IT Service Management (ITSM) because it helps organizations track issues, manage workflows, and improve service delivery across different departments.ServiceNow operates using a table-based architecture, where all information such as users, incidents, groups, and tasks are stored as records in database tables. These tables are connected through reference fields, allowing the system to maintain relationships between different types of data. For example, a support ticket can be linked to a specific user or support group responsible for resolving the issue.

The platform also offers powerful automation tools such as Flow Designer, which allows organizations to create workflows that automatically perform actions when certain conditions are met. This helps reduce manual effort, improves efficiency, and ensures that tasks such as ticket assignment, notifications, and approvals are handled quickly and accurately. Overall, ServiceNow helps organizations streamline operations, improve productivity, and deliver better support services.

2.2 Importance of Automated Ticket Assignment

In organizations that offer multiple products and services, customer support teams receive a large number of support tickets every day. These tickets may include issues such as login problems, system errors, access requests, or service-related queries. When tickets are assigned manually, support staff must review each request and determine the appropriate team responsible for resolving the issue. This process can be time-consuming and may lead to several operational challenges such as delays in ticket resolution, incorrect ticket assignment, increased workload for support staff, and reduced customer satisfaction.

Automated ticket assignment helps overcome these challenges by using predefined rules and conditions to automatically route tickets to the correct support group. By analyzing the issue type or category of the ticket, the system immediately assigns it to the appropriate team without manual intervention. This ensures that the ticket reaches the right department quickly, allowing support teams to begin resolving the issue faster. As a result, automation improves response time, reduces human errors, enhances operational efficiency, and provides a better support experience for customers.

3. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To automate the process of ticket assignment in ServiceNow.
- To create users, groups, and roles for managing support teams.
- To design a custom table for handling operational issues.
- To configure access control for secure data management.
- To implement automated flows for routing tickets to appropriate groups.
- To improve operational efficiency and customer satisfaction.
- To reduce manual effort in ticket management.

4. TOOLS AND TECHNOLOGIES USED

Tool / Technology	**Purpose**
ServiceNow Platform	Used to build and automate the ticket management system
Flow Designer	Used to automate ticket assignment workflows
Users and Groups	Used to manage support team members
Roles	Used to control system access
Access Control Lists (ACL)	Used to secure system data
Custom Tables	Used to store ticket-related information

5. SYSTEM ARCHITECTURE /WORKFLOW

User Request / Ticket Creation → Operations Related Table → Flow Designer Trigger → Issue Classification → Automatic Group Assignment → Ticket Resolution.

Step	Stage	Component	Key Activity & Impact
1. Ticket Creation & Storage	Stage 1 & 2	User / Support Ticket Operations Related Table	**Activity:** A user submits a ticket describing an issue (e.g., login failure). This ticket is stored in a custom ServiceNow table **Impact:** Captures critical data and serves as the initial system input.
2. Automated Classification & Assignment	Stage 3, 4, & 5	Flow Designer Trigger Issue Classification Automatic Ticket Assignment	**Activity:** A Flow Designer workflow triggers, classifies the issue category (e.g., 404 error), and automatically assigns the ticket to the appropriate group (e.g., Platform Support). **Impact:** Ensures immediate categorization and speeds up the time to the correct technician.
3. Resolution & Closure	Stage 6	Support Team Resolution	**Activity:** The assigned support group reviews the ticket, resolves the issue, and updates the ticket status **Impact:** Leads to resolution, a complete audit trail, and faster overall support operation efficiency.

Workflow Explanation

Milestone 1 — Ticket Creation and Storage: Users create support tickets and the details are stored in the Operations Related table in ServiceNow

Milestone 2 — Automated Ticket Routing: Flow Designer automatically analyzes the issue type and assigns the ticket to the appropriate support group such as Platform

6. PROJECT IMPLEMENTATION STEPS

The implementation of the project was carried out in a structured sequence of steps as described below:

Step 1: Creating Users

The first step in the implementation process involved creating users in the ServiceNow platform. Users represent individuals who interact with the system, such as support staff responsible for resolving tickets. The required user details such as name, email address, and role were entered while creating the user records. These users were later assigned to specific support groups that handle different types of issues.

Step 2: Creating Support Groups

After creating users, support groups were created to represent teams responsible for resolving different types of tickets. For this project, two groups were created: Platform Support Group and Certificates Support Group. Each group consists of users who are responsible for resolving issues related to their specific area. These groups help organize support staff and simplify the ticket assignment process.

Step 3: Creating Roles

Roles were created in the system to define access permissions for users. Roles ensure that only authorized users can access or modify certain records in the system. In this project, roles such as Platform Role and Certificate Role were created. These roles were later assigned to the respective users and groups to maintain proper security and access control within the system.

Step 4: Creating a Custom Table

A custom table named Operations Related was created in the ServiceNow platform to store support ticket information. This table acts as the primary location where all support requests are recorded. Fields such as Issue Type, Description, and Assigned Group were created in this table. The Issue Type field contains options such as login issues, platform errors, certificate requests, and user expiration issues.

Step 5: Assigning Users and Roles to Groups

Once the users, groups, and roles were created, users were assigned to the appropriate groups and roles. For example, users responsible for handling certificate-related issues were assigned to the Certificates Group, while users responsible for platform-related issues were assigned to the Platform Group. This step ensured that each support group had the necessary members to resolve assigned tickets.

Step 6: Configuring Access Control Lists (ACL)

Access Control Lists (ACLs) were configured to manage security and control access to the custom table. ACL rules were created to ensure that only authorized users with specific roles could read, create, update, or delete records in the Operations Related table. This helped protect sensitive data and maintain proper system security.

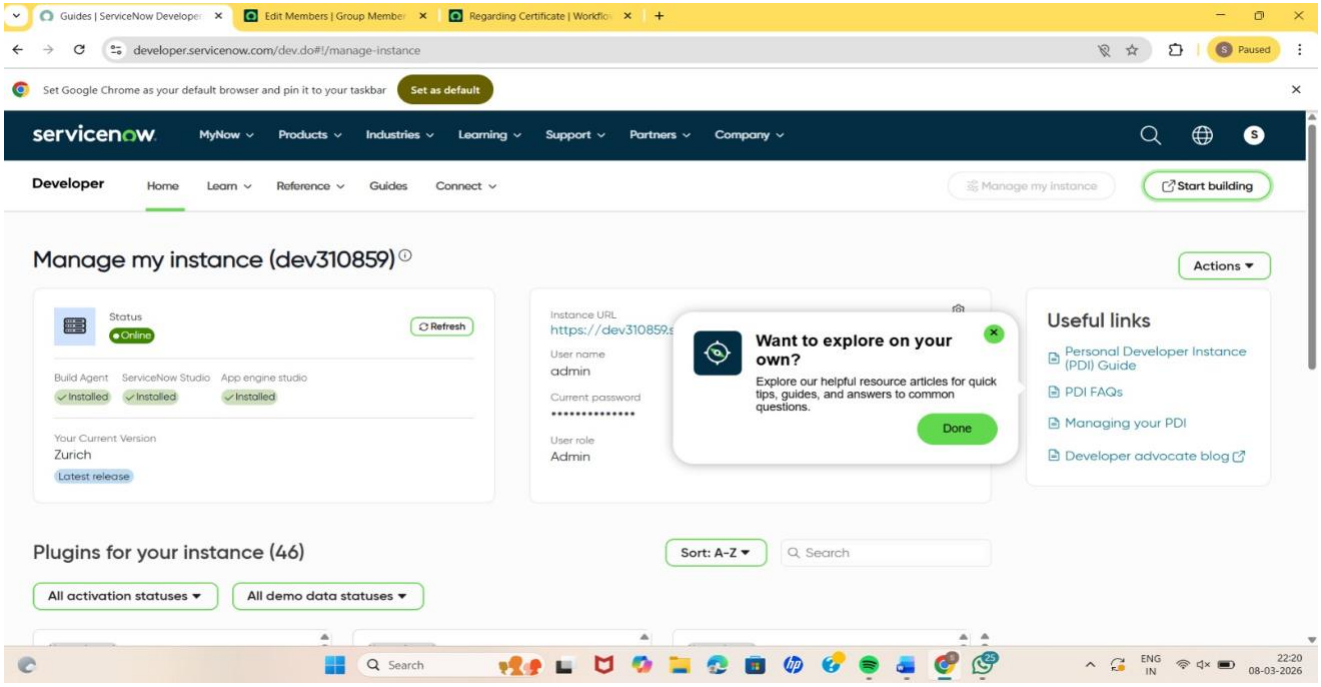
Step 7: Creating Automated Flows using Flow Designer

Automation was implemented using Flow Designer, which allows workflows to be created through a visual interface. Two flows were created in this project. The first flow handles certificate-related issues, while the second flow handles platform-related issues. These flows are triggered whenever a new ticket is created in the Operations Related table.

Step 8: Running the Automated Ticket Assignment Process

When a ticket is created, the Flow Designer workflow automatically checks the Issue Type field. Based on the selected issue category, the system assigns the ticket to the appropriate support group. For example, if the issue is related to certificates, the ticket is assigned to the Certificates Group, and if the issue is related to platform errors or login problems, the ticket is assigned to the Platform Group. This automated process ensures accurate ticket routing and improves support efficiency.

7.SCREENSHOTS



Guides | ServiceNow Developer | Groups | ServiceNow | Regarding Certificate | Workflow

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Set Google Chrome as your default browser and pin it to your taskbar | Set as default

servicenow | All | Favorites | History | Workspaces | Admin | Groups | Search

Groups | Name | Search | Actions on selected rows... | New

All > Name >= platform

Name	Description	Active	Manager	Parent	Updated
platform		true	(empty)	(empty)	2026-03-08 07:27:35
Problem Analyzers	Group that works on analyzing and fixing...	true	(empty)	(empty)	2026-01-04 21:11:33
Problem Solving	Group that works on analyzing and fixing...	true	(empty)	(empty)	2026-01-04 21:10:15
Procurement	Responsible for ordering catalog items, ...	true	(empty)	(empty)	2021-06-11 18:50:16
Project Mgmt	Responsible for Return Material Authorization approvals and RMA number allocations in Inventory Management	true	(empty)	(empty)	2024-05-09 18:42:19
Recommendation Admin		true	(empty)	(empty)	2019-02-14 16:39:04
Report Access Request Approvers		true	(empty)	(empty)	2021-03-15 11:50:37
RMA Approvers	Responsible for Return Material Authoriz...	true	(empty)	(empty)	2026-01-05 13:12:50
Service Desk		true	Beth Anglin	(empty)	2026-01-04 16:14:56
Software	IT department responsible for all softwa...	true	(empty)	(empty)	2021-06-11 18:53:27
Survey Creators	This is a group of users who have survey...	true	(empty)	(empty)	2019-02-11 11:36:56
Team Development Code Reviewers	Review, approve and/or reject the code p...	true	(empty)	(empty)	2013-12-01 15:47:36
US Presidents Group 1	Demo user group for Password Reset Appli...	true	(empty)	(empty)	2013-07-25 15:20:25
US Presidents Group 2	Demo user group for Password Reset Appli...	true	(empty)	(empty)	2013-07-25 15:21:07

1 to 14 of 14

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Guides | ServiceNow Developer | Groups | ServiceNow | Regarding Certificate | Workflow

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Set Google Chrome as your default browser and pin it to your taskbar | Set as default

servicenow | All | Favorites | History | Workspaces | Admin | Groups | Search

Groups | Name | Search | Actions on selected rows... | New

All > Name >= certificates

Name	Description	Active	Manager	Parent	Updated
certificates		true	(empty)	(empty)	2026-03-08 07:26:59
Change Management	Change Management Group	true	(empty)	(empty)	2015-06-09 22:35:12
Conditional Script Writer	This is the default group introduced by ...	true	(empty)	(empty)	2025-02-25 18:04:32
Creator Studio Restricted Users	Provides access to the Creator Studio de...	true	(empty)	(empty)	2023-12-07 15:58:22
Creator Studio Users	Provides access to the Creator Studio de...	true	(empty)	(empty)	2023-12-07 15:48:42
Database		true	Don Goodliffe	(empty)	2023-03-13 19:02:18
Database Atlanta	Manages databases hosted in Atlanta	true	(empty)	Database	2024-03-30 11:47:18
Database San Diego	Manages databases hosted in San Diego	true	(empty)	Database	2024-03-30 11:46:37
eCAB Approval	eCAB Approvers	true	(empty)	(empty)	2011-09-30 01:46:44
Field Services		true	(empty)	(empty)	2021-06-07 18:11:34
Hardware	IT department responsible for all hardwa...	true	(empty)	(empty)	2021-06-11 18:54:14
Help Desk		true	ATF Change Management	(empty)	2026-01-12 01:50:32
Incident Management	Incident Management Group	true	(empty)	(empty)	2017-05-16 04:13:49
IT Finance CAB		true	(empty)	(empty)	2021-10-30 14:00:17

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Guides | ServiceNow Developer | Edit Members | Group Role | Se | Regarding Certificate | Workflo

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servicenow All Favorites History Workspaces **Group Role - Edit Members** Search

Edit Members **Cancel** **Save**

Add Filter **Run filter**

-- choose field -- | -- oper -- | -- value --

Collection

Roles List

certificates

access_analyzer_admin
access_analyzer_ai_user
action_category_creator
action_designer
action_write_enabled
activity_admin
activity_creator
actsub_admin
actsub_user
admin
agent_admin
agent_security_admin
agent_workspace_user
ai_native_experience_analytics_admin
ai_native_experience_analytics_viewer

Certification_role

Cancel **Save**

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Guides | ServiceNow Developer | action_designer | Role | Service | Regarding Certificate | Workflo

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Set Google Chrome as your default browser and pin it to your taskbar **Set as default**

servicenow All Favorites History Workspaces Admin **Role - action_designer** Search

Role
action_designer

Update **Delet**

Name: action_designer

Application: Global

Elevated privilege: ☐

Description: action designer role enables users to launch Action Designer

Update **Delete**

Contains Roles (2) | Applications with Role (1) | Modules with Role (8) | Custom Tables

for text | Search | Actions on selected rows... | **New** | **Edit...**

Role = action_designer

Contains
action_write_enabled
sn_workflow_studio.workflow_studio_read

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Guides | ServiceNow Developer | certificates | Group | ServiceNo | Regarding Certificate | Workflo

dev310859.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D659e4b70c3277210ea57befdd401310b%26

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

Workflow Studio | Regarding Certificate Flow • Global | Regarding Platform Flow • Global

Regarding Certificate Active

View: | Test | Edit flow | Deactivate

TRIGGER

Operations related Created or Updated where (Issue is regarding certificates)

ACTIONS

1 Update Operations related Record

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data Expand All

- Flow Variables
- Trigger - Record Created or Updated
- 1 - Update Record

Read-only | Status: Published | Application: Global

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Guides | ServiceNow Developer | certificates | Group | ServiceNo | Regarding Platform | Workflow

dev310859.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D03929b38c3277210ea57befdd40131e5%26

Set Google Chrome as your default browser and pin it to your taskbar | Set as default

Workflow Studio | Regarding Certificate Flow • Global | Regarding Platform Flow • Global

Regarding Platform Active

View: | Test | Edit flow | Deactivate

TRIGGER

Operations related Created or Updated where (Issue is unable to login to platform; Issue is 404 error; Issue is regarding user expired)

ACTIONS

1 Update Operations related Record

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data Expand All

- Flow Variables
- Trigger - Record Created or Updated
- 1 - Update Record

Read-only | Status: Published | Application: Global

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8. RESULTS AND OUTPUT

After implementing the automated ticket assignment system in the ServiceNow platform, several positive outcomes were observed. The system successfully improved the efficiency of the support process and ensured that tickets were handled in a more organized and faster manner. The following results were achieved:

- Tickets were automatically assigned to the correct support groups based on the issue type.
- Manual ticket routing by administrators or support managers was eliminated.
- Support teams received tickets immediately after they were created in the system.
- The chances of incorrect ticket assignment were significantly reduced.
- Issue resolution time improved because tickets reached the right team without delay.
- The overall workflow of the support process became more structured and efficient.
- Support staff were able to focus more on resolving issues instead of managing ticket distribution.
- The system ensured better coordination between different support teams.
- The automated process reduced human errors that commonly occur during manual assignment.
- Ticket tracking and management became easier through the centralized ServiceNow platform.
- The workload among support teams became more balanced due to proper routing of tickets.
- Customer satisfaction increased because issues were addressed more quickly and accurately.
- The automation improved the overall operational efficiency of the support department.
- The system demonstrated how workflow automation can enhance service management processes in organizations.

9. ADVANTAGES

The implementation of the automated ticket assignment system provides several benefits to the support operations. These advantages help improve efficiency, accuracy, and overall service quality within the organization.

Faster Ticket Assignment:

The system automatically assigns tickets to the appropriate support group as soon as they are created. This eliminates delays that occur in manual routing and ensures that issues are addressed quickly.

Reduced Manual Effort:

Automation removes the need for support managers or administrators to manually review and assign tickets. This reduces the workload on staff and allows them to focus on resolving issues rather than managing ticket distribution.

Improved Customer Satisfaction:

Since tickets are routed instantly to the correct support team, customer issues are resolved more quickly. Faster response and resolution times lead to a better support experience for customers.

Efficient Resource Utilization:

The system ensures that tickets are assigned to the correct teams based on the type of issue. This helps utilize the skills and expertise of support staff effectively and prevents unnecessary workload on unrelated teams.

Better Ticket Tracking:

All tickets are stored and managed within the ServiceNow platform, allowing easy tracking of ticket status, assignment, and resolution progress. This improves transparency and accountability in support operations.

Increased Productivity:

. By reducing manual tasks and improving workflow efficiency, support teams can handle more tickets in less time. This increases the overall productivity of the support department and improves service performance.

10.CONCLUSION

The implementation of the automated ticket routing system significantly improved the efficiency of support operations at ABC Corporation. By using ServiceNow features such as users, groups, roles, Access Control Lists (ACL), and Flow Designer, the process of assigning support tickets to the appropriate teams was successfully automated. This reduced the need for manual ticket routing and ensured that each support request was directed to the correct team based on the issue type.

The automated system helped reduce delays in ticket assignment and minimized the chances of incorrect routing. As a result, support teams were able to receive tickets faster and begin resolving issues without unnecessary waiting time. This improved the overall workflow of the support process and allowed support staff to focus more on resolving problems rather than managing ticket distribution.

Another important outcome of the system was improved operational efficiency. With automation handling repetitive tasks, the workload on support administrators was reduced, and the process became more organized and reliable. The centralized system also made it easier to track, monitor, and manage support tickets within the organization.

Additionally, the faster response time and accurate ticket routing contributed to higher customer satisfaction. Users were able to receive quicker support for their issues, which improved their overall experience with the organization's services.

In conclusion, the project demonstrates how workflow automation using the ServiceNow platform can streamline support operations, reduce manual effort, and improve service quality. The automated ticket assignment system provides an effective solution for managing support requests and can be further enhanced in the future with advanced features to improve IT service management processes.

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Project Design Phase

The Project Design Phase is the stage where the overall structure and design of the system are planned before development begins. In this phase, developers create the system architecture, database design, and user interface layout to ensure the system works efficiently.For the Streamlining Ticket Assignment System, the design phase focuses on creating a clear structure for how tickets will be created, processed, assigned, and tracked in the system.The main objective of this phase is to design a system that is easy to use, efficient, and reliable for handling support tickets.

System Architecture Design

- The system architecture defines how different parts of the application interact with each other.
- The ticket assignment system usually follows a three-layer architecture:

1. Presentation Layer

- This is the user interface where users interact with the system.
- It includes login pages, ticket creation forms, and dashboards.

2. Application Layer

- This layer contains the business logic of the system.
- It processes ticket information and performs automatic assignment.

3. Database Layer

- This layer stores all the system data such as users, tickets, and departments.
- This architecture helps in maintaining the system easily and improving performance.

User Interface Design

The user interface is designed to make the system simple and easy to use.

Main Screens:

- Login Page
- Dashboard
- Ticket Creation Form
- Ticket Status Page
- Admin Management Panel

Example:

A customer opens the Ticket Creation Form, enters the problem description, and submits the ticket.

4. Ticket Assignment Logic Design

The design phase also defines the rules used to assign tickets automatically.

Assignment Logic:

- If the issue category is Software, assign it to the Software Support Team.
- If the issue category is Hardware, assign it to the Hardware Support Team.
- If the issue category is Network, assign it to the Network Support Team.

This rule-based design helps the system assign tickets quickly without manual intervention

5. Data Flow Design

The Data Flow shows how information moves through the system.

Example Flow:

1. User creates a ticket.

2. Ticket data is sent to the application layer.

3. The system checks the issue category.

4. The ticket is automatically assigned to the correct support agent.

5. The database stores the ticket information.

6. The agent receives the assigned ticket and starts working on it.

Example Scenario

A customer submits a ticket with the issue “Laptop keyboard is not working.”

1. The ticket is created in the system.

2. The system checks the category and identifies it as a hardware issue.

3. According to the assignment rules, the ticket is assigned to the Hardware Support Team.

4. The support agent receives the ticket and updates its status to In Progress.

Through proper design, the Streamlining Ticket Assignment System ensures that tickets are handled quickly and efficiently.

Requirement Analysis Phase

The Requirement Analysis Phase is the stage where all the needs of the system are collected and analyzed before developing the application. In this phase, developers understand what features the system should provide and how users will interact with it. For the Ticket Assignment System, the requirement analysis focuses on improving how support tickets are created, managed, and automatically assigned to the correct department or support agent.

The system should:

- Allow users to create support tickets easily.
- Automatically assign tickets to the correct department or manager.
- Allow support staff to view, update, and resolve tickets.
- Provide tracking and status updates for each ticket.
- Store all ticket information in a database for future reference.

By analyzing these requirements, the organization can design a system that reduces manual work and improves support efficiency.

Introduction

Customer support systems are important for organizations to handle customer issues quickly and effectively. When customers face problems with products or services, they submit a support request which becomes a ticket in the system.In many organizations, ticket assignment is done manually, which can cause delays and errors. The Streamlining Ticket Assignment System helps solve this problem by automatically assigning tickets to the appropriate department or support staff based on the issue type.

This system improves:

- Response time
- Workload distribution
- Customer satisfaction
- Operational efficiency

User Roles

The system includes different users, and each user has specific responsibilities.

1.Admin

The Admin manages the entire ticket system.

Responsibilities:

- Create and manage departments.

- Add or remove support staff.
- Set rules for automatic ticket assignment.
- Monitor ticket status and reports.

Example:

If a ticket is related to a network issue, the admin can configure the system so that it is automatically assigned to the Network Support Team.

2.Support Staff / Agent

Support staff members are responsible for resolving customer issues.

Responsibilities:

- View assigned tickets.
- Update ticket status (Open, In Progress, Resolved).
- Communicate with customers.
- Provide solutions for reported problems.

Example:

A Mobile App Support Agent receives a ticket about an app crash and works to fix the problem.

3.User / Customer

The user or customer creates support tickets when they face issues.

Responsibilities:

- Create new support tickets.
- Provide issue details.
- Track ticket status.
- Receive updates and solutions.

Example:

A customer submits a ticket saying “The mobile app is not loading after login.”

Libraries Role

Libraries are software components that help developers build the system easily by providing ready-made functions.

Examples of libraries used in the system:

1.UI Libraries

- Used to design the user interface of the application.

- Helps create forms, buttons, and dashboards.

2.Authentication Libraries

- Used for login and user authentication.
- Ensures secure access for admin, staff, and users.

3.Database Libraries

- Helps connect the application to the database.
- Used for storing and retrieving ticket data.

4.Notification Libraries

- Used to send notifications when a ticket is created or updated.

Example:

When a user submits a ticket, a notification library can send a message to the assigned support staff.

Database Tables

The system stores all information in a database using tables.

1.Users Table

Stores details about users and support staff.

2.Tickets Table

Stores all ticket details.

3.Departments Table

Stores information about departments.

4.Ticket Assignment Table

Stores information about ticket assignments.

Example Scenario

1. A customer submits a ticket saying “My mobile app is not working.”
2. The system analyzes the issue type and identifies it as a software problem.
3. The ticket is automatically assigned to the Software Support Agent.
4. The agent views the ticket, updates the status to In Progress, and later marks it Resolved after fixing the issue.

This automated process makes support operations faster and more efficient.

Identification Phase

The Identification Phase is the first stage where the problem in the current system is identified and clearly understood.

In many organizations, customer support teams receive a large number of support tickets every day. These tickets may include issues related to technical problems, service requests, or general inquiries. If the ticket assignment process is done manually, it can create several issues such as delays, incorrect assignment, and increased workload for managers.

During the identification phase, the organization studies the existing ticket management system and identifies the difficulties faced by support staff. The main goal is to find out why ticket handling is slow or inefficient.

Introduction

Customer support plays an important role in maintaining customer satisfaction and service quality. Organizations receive support requests through emails, applications, or helpdesk systems, which are converted into tickets.Managing these tickets efficiently is essential to ensure that customer issues are resolved quickly. However, when the ticket assignment process is not organized, it can lead to slow responses, confusion among staff, and poor customer experience.The concept of Streamlining Ticket Assignment for Efficient Support Operations focuses on improving the process of assigning support tickets to the right department or support agent automatically. This helps reduce manual effort, improve response time, and ensure that tickets are handled by the most suitable person.

An efficient ticket assignment system ensures:

- Faster issue resolution
- Proper workload distribution
- Improved productivity of support teams
- Better customer satisfaction

Problem Statement

In many support systems, ticket assignment is handled manually by administrators or managers. This process can cause delays and mistakes, especially when there are many incoming tickets.For example, consider an IT support company that receives tickets related to software issues, hardware issues, and network problems. If all tickets first go to a manager who manually assigns them, the process becomes slow. Sometimes a software issue ticket may be mistakenly assigned to a hardware team, which delays the solution.

Example:

- A customer reports that their mobile application is not working and submits a ticket.
- The ticket should ideally go to the Mobile Application Support Team.
- But if the assignment is manual, it may first go to the General Support Team.
- The ticket then has to be reassigned, which wastes time and delays the resolution.

Because of this problem:

- Customers wait longer for solutions.
- Support staff spend more time managing tickets rather than solving issues.
- The organization's efficiency decreases.

Therefore, a streamlined ticket assignment system is needed to automatically route tickets to the correct department and manager based on the issue type, ensuring faster and more efficient support operations.

Project planning phase

The Project Planning Phase is the stage where the development team plans how the project will be executed. In this phase, the team defines the project scope, resources, timeline, and tasks required to successfully develop the system.For the Streamlining Ticket Assignment System, the project planning phase focuses on organizing the development process so that the system can efficiently manage and automatically assign support tickets.The main goal of this phase is to ensure that the project is completed on time, within budget, and with the required quality

1.Project Objectives

The main objectives of the project are:

- To develop a ticket management system that handles customer support requests.
- To automatically assign tickets to the appropriate department or support agent.
- To reduce manual work and delays in ticket assignment.
- To improve support team productivity and customer satisfaction.

2. Project Scope

The project scope defines what features will be included in the system.

Included in the project:

- User registration and login system
- Ticket creation by customers
- Automatic ticket assignment to departments or agents
- Ticket status tracking (Open, In Progress, Resolved)
- Admin dashboard for managing users and tickets

Not included in the project:

- External customer communication tools
- Advanced AI-based ticket analysis (for future development)

3. Resource Planning

Resources required for the project include both human resources and technical resources.

Human Resources

- Project Manager – Oversees the entire project
- System Analyst – Analyzes system requirements
- Developers – Build the application
- Database Administrator – Manages the database
- Testing Team – Tests the system for errors

Technical Resources

- Computer systems
- Development software and tools
- Database management system
- Internet and server support

4. Risk Management

During project planning, possible risks are identified and solutions are prepared.

Possible Risks:

Delay in development

Technical errors in ticket assignment

Database issues

Solutions:

- Regular progress monitoring
- Proper testing of the system
- Backup and recovery plans for data

5.Example Scenario

Suppose an organization receives hundreds of support tickets daily. Without proper planning, the development of a ticket assignment system may face delays or incomplete features.By following the project planning phase, the team clearly defines tasks, assigns responsibilities, and sets deadlines. As a result, the system is developed smoothly and helps automatically assign tickets to the correct support team, improving the efficiency of support operations.

Project Development Phase

The Project Development Phase is the stage where the actual system is built based on the requirements and design created in the previous phases. In this phase, developers write the code, create the database, and develop the user interface for the ticket assignment system.For the Streamlining Ticket Assignment System, the development phase focuses on creating modules that help users create tickets, automatically assign them to the correct department or support agent, and track the progress of each ticket.The main goal of this phase is to develop a full functional system that can manage support tickets efficiently.

1. System Design and Architecture

During development, the system architecture is created to define how different parts of the application work together.

The system generally follows a three-tier architecture:

1. Presentation Layer – User interface where users create and view tickets.

2. Application Layer – Processes ticket data and applies assignment rules.

3. Database Layer – Stores all ticket and user information.

This structure helps in organizing the system and improving performance.

2. Module Development

The project is divided into different modules, and each module is developed separately.

1. User Management Module

This module manages user accounts such as Admin, Support Staff, and Customers.

Functions:

- User registration
- Login and authentication
- Role management

Example:

A new support agent account is created by the admin.

2. Ticket Creation Module

This module allows customers or users to create support tickets.

Functions:

- Enter ticket title and description
- Select issue category
- Submit the ticket to the system

Example:

A customer submits a ticket saying “My mobile application crashes during login.”

3. Ticket Assignment Module

This is the main feature of the system. It automatically assigns tickets to the correct department or support agent.

Functions:

- Identify issue type
- Match ticket with appropriate department
- Assign ticket to available support staff

Example:

A software-related issue is automatically assigned to the Software Support Team.

4. Ticket Tracking Module

This module allows users and staff to monitor the status of tickets.

Ticket status may include:

- Open
- In Progress
- Resolved
- Closed

Example:

A support agent changes the ticket status from Open to In Progress when they start working on it.

5. Notification Module

This module sends notifications to users and staff when a ticket is created, assigned, or updated.

Example:

- A support agent receives a notification when a new ticket is assigned.
- The customer receives an update when the issue is resolved.

3. Database Development

During this phase, database tables are created to store system information such as:

User details

Ticket information

Department details

Ticket assignment records

The database ensures that all data is stored securely and can be accessed when needed.

4. Integration of Modules

After developing all modules, they are integrated to ensure the system works as a complete application.

5. Example Scenario

1. A customer logs into the system and creates a ticket:

"Internet connection is not working."

2. The system analyzes the issue and identifies it as a network problem.

3. The ticket is automatically assigned to the Network Support Team.

4. The assigned support agent receives a notification and updates the ticket status to In Progress.

5. After solving the issue, the agent marks the ticket as Resolved, and the customer is notified.

Through this development process, the Streamlining Ticket Assignment System helps organizations handle support requests faster and more efficiently.

Project Documentation Phase

The Project Documentation Phase is the stage where all the information about the project is recorded and organized in a clear format. Documentation helps developers, administrators, and users understand how the system works and how it should be used.For the Streamlining Ticket Assignment System, documentation includes details about the system design, development process, database structure, and user instructions. Proper documentation makes it easier to maintain, update, and use the system in the future.The main purpose of this phase is to ensure that all project details are well documented, easy to understand, and available for future reference.

1. System Documentation

System documentation explains the technical details of the project. It is mainly used by developers and system administrators.

It includes:

- System architecture and design
- Database structure
- Modules and their functions
- Technologies and tools used
- System workflow

Example:

Documentation describes how the ticket assignment module automatically assigns tickets to the correct department based on issue category.

2. User Documentation

User documentation explains how users can operate the system. It provides step-by-step instructions for different users such as Admin, Support Staff, and Customers.

It includes:

- How to log in to the system
- How to create a support ticket
- How to check ticket status
- How support agents update ticket progress

Example:

A customer guide explains how to create a ticket by entering the issue title, description, and category.

3. Technical Documentation

Technical documentation contains detailed information for developers who may maintain or upgrade the system in the future.

It includes:

- Source code explanation
- API or system integration details
- Software libraries used
- Database tables and relationships

Example:

The documentation explains how the system connects to the database and stores ticket information.

4. Testing Documentation

Testing documentation records the results of system testing. It ensures that the system works correctly before deployment.

It includes:

- Test cases
- Expected results
- Actual results
- Bug reports and fixes

Example:

A test case checks whether a software-related ticket is correctly assigned to the Software Support Team.

5. Maintenance Documentation

Maintenance documentation helps future developers maintain and update the system.

It includes:

- System update procedures
- Backup and recovery methods
- Troubleshooting steps
- Future improvement suggestions

Example:

Instructions are provided for adding new departments or support agents in the system.

Example Scenario

- Suppose a new support staff member joins the company. By using the project documentation, they can easily learn how to log in, view assigned tickets, update ticket status, and resolve customer issues.
- Similarly, if developers need to improve the system later, they can refer to the technical documentation to understand the database structure and system modules.
- Thus, the Project Documentation Phase ensures that the Streamlining Ticket Assignment System is easy to understand, use, and maintain in the future.

Project Demonstration Phase

The Project Demonstration Phase is the final stage of the project where the developed system is presented to users, instructors, or stakeholders to show how it works. In this phase, the main features of the system are demonstrated to prove that the project meets the requirements and works correctly.For the Streamlining Ticket Assignment System, the demonstration shows how tickets are created, automatically assigned to the correct support team, and resolved efficiently.

The purpose of this phase is to:

- Show the working model of the system
- Explain the features and modules
- Demonstrate how the system improves support operations
- Collect feedback and suggestions

1. System Overview Demonstration

First, the overall structure of the system is explained. The presenter introduces the main components of the system such as:

- User login and authentication
- Ticket creation process
- Automatic ticket assignment
- Ticket tracking and status updates
- Admin management features

This helps the audience understand the purpose and functionality of the system.

2. User Login Demonstration

In this step, the login process is demonstrated.

Example:

- Admin logs into the system using username and password.
- Support agents and customers can also log in with their credentials.
- After login, users are redirected to their respective dashboards.

3. Ticket Creation Demonstration

Next, the process of creating a support ticket is shown.

Example:

1. A customer logs into the system.

2. The customer clicks “Create Ticket.”

3. The user enters the problem details such as:

Ticket Title: Mobile App Not Working

Description: The application crashes during login.

Category: Software Issue

After submitting the form, the ticket is saved in the system.

4. Automatic Ticket Assignment Demonstration

This is the main feature of the project.

The system automatically assigns the ticket to the appropriate department or support staff based on the issue category.

Example:

- If the issue is related to software, the ticket is assigned to the Software Support Team.
- If the issue is related to network problems, it is assigned to the Network Support Team.
- This automatic process reduces manual effort and saves time.

5. Ticket Tracking Demonstration

The demonstration also shows how tickets are tracked.

Support staff can:

- View assigned tickets
- Update ticket status
- Add comments or solutions

Example:

- Status changes from Open → In Progress → Resolved.
- Customers can also check the progress of their tickets.

6. Admin Dashboard Demonstration

The Admin Dashboard is also demonstrated to show management features.

Admin can:

- Manage users and support agents
- Create departments
- Monitor ticket statistics
- View reports and system performance

This helps the organization manage support operations efficiently.

7. Example Demonstration Scenario

1. A customer reports a problem: “Internet connection is not working.”

2. The customer creates a ticket in the system.

3. The system automatically assigns the ticket to the Network Support Team.

4. The support agent receives the ticket and changes the status to In Progress.

5. After fixing the problem, the agent updates the ticket status to Resolved.

6. The customer receives a notification that the issue has been solved.

Conclusion of Demonstration

The demonstration shows that the Streamlining Ticket Assignment System successfully improves the ticket handling process. It reduces manual work, assigns tickets quickly, and helps support teams resolve customer issues more efficiently.